

CANONICAL LESSON

# From Signal to Accountable Action

Decision-centred foundations for accountable sector intelligence

|                 |                                   |
|-----------------|-----------------------------------|
| Production Unit | PU-B07-C05                        |
| Notebook        | V7_0_N05                          |
| Status          | Controlled educational draft v0.1 |
| Data boundary   | Synthetic instructional data only |

**Central decision:** Convert analytical signals into capacity-aware, reviewable, monitored, and retireable decision products.

## Learning outcomes

- Transform analytical signals into governed decision products: scenarios, intervals, reason codes, dashboards, alerts, human review, monitoring, incident response, and retirement.

## Why this matters

An unreviewable queue or unexplained dashboard score creates operational noise and can shift authority silently to the model.

The analytical workflow is useful only when evidence, uncertainty, authority, and action are connected explicitly. The model supports professional judgement; it does not replace accountable institutional authority.

### 1. Separate signal from decision

- A score becomes a review disposition only after quality, uncertainty, capacity, authority, and escalation rules are applied.

### 2. Capacity-aware queues

- An alert that cannot be reviewed is not actionable.
- Prioritization should expose unmet need rather than hide it.

### 3. Executive dashboard contract

- Every card needs a definition, reference period, geography, source vintage, uncertainty/status, owner, action, and link to evidence.

### 4. Alert deduplication and escalation

- Repeated signals should update one case/event rather than create alert fatigue.
- Escalation is role-based and time-bound.

### 5. Monitoring after deployment

- Monitor data quality, coverage/calibration where targets mature, workload, overrides, unresolved cases, incidents, and subgroup/geographic effects.

### 6. Incident and retirement controls

- A material breach triggers containment, notification, investigation, correction, validation, and documented restart—or retirement.

### 7. Decision brief

- Executives need the decision, evidence, uncertainty, affected groups, options, recommendation, owner, deadline, safeguards, and follow-up—not an unexplained model score.

### Worked cross-sector example

Repeated alerts update one governed case with acknowledgement, escalation, resolution, and expiry times rather than creating alert fatigue.

**Boundary:** The example supports learning and review. It does not constitute an official statistic, eligibility determination, enforcement action, clinical judgement, or operational authorization.

### Practice and reflection

- 1.
- Add alert expiry and acknowledgement.
- 2.
- Define an override review.
- 3.
- Distinguish monitoring from impact evaluation.
- 4.
- Draft a retirement record.

## Answer key

- 1.
- Store created, acknowledged, resolved, and expiry times with escalation on missed service levels.
- 2.
- Review rates, reasons, outcomes, users, sectors/groups, and whether rules or training need change.
- 3.
- Monitoring checks operation and model behaviour; impact evaluation estimates whether the service caused better outcomes.
- 4.
- Record reason, authority, effective date, affected users, replacement, data/model disposition, retained evidence, and communications.

## Operational readiness checklist

- Purpose and decision owner are named.
- Evidence definitions, provenance, and lawful access are documented.
- Uncertainty and abstention are visible.
- Human review, escalation, monitoring, incident response, and retirement are assigned.
- Equity, accessibility, security, and sector-specific approval are complete.